

RESEARCH

Open Access



Advancing sustainable tourism in Lebanon through hotel sustainability practices and stakeholder collaboration toward achieving sustainable development goals

Nadia Kabbara^{1*}, Bassel El Hajj^{2†} and Iman Baba^{3†}

[†]Bassel El Hajj and Iman Baba contributed equally to this work.

*Correspondence:

Nadia Kabbara
nadia.mkabbara@gmail.com

¹School of Business, Lebanese International University, Tripoli, Lebanon

²Faculty of Business, University of Tripoli, Tripoli, Lebanon

³Faculty of Tourism, City University, Tripoli, Lebanon

Abstract

Tourism plays a significant role in Lebanon's economic growth. However, the sector faces persistent hurdles linked to political instability, economic crises, and the COVID-19 pandemic. Promoting sustainable tourism offers a powerful driver to prosperity and alignment with the Sustainable Development Goals (SDGs). This study examines sustainability practices within Lebanon's hotel sector, exploring key constraints and opportunities for improvement. Qualitative data were collected through unstructured interviews with 15 hoteliers across Lebanon and analyzed using thematic analysis. Findings reveal that hotels are implementing various economic, social, and environmental practices but continue to face significant challenges related to resource constraints, policy gaps, and stakeholder coordination. The study highlights the need for stronger collaboration among stakeholders and more integrated sustainability strategies to enhance Lebanon's tourism sector. Insights from this research can inform policymakers and industry leaders in developing effective approaches to achieve sustainable tourism and long-term sectoral resilience.

Keywords Sustainable tourism, Economic sustainability, Social sustainability, Environmental sustainability, Green initiatives, Green HR, Sustainable development goals (SDGs)

1 Introduction

Sustainable Development Goals (SDGs), launched by the United Nations in 2015, seek to achieve a better and more sustainable future for all by addressing the global challenges the world is facing [110]. Meeting these goals, according to the United Nations Global Compact [109], requires an extraordinary effort from all sectors of society, with a particular focus on the important role that businesses play in the process. The tourism and hospitality industry are one of the main players in the global economy and has extensive environmental and social impacts [47]. The World Tourism Organization and the United



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Nations Global Compact Network Spain [119] emphasized the importance of this sector in contributing to the successful achievement of the SDGs. Recent international studies continue to show that tourism is essential for advancing the achievement of SDGs particularly in developing economies challenged with structural constraints [56].

Lebanon established a successful tourism development policy and experienced high prosperity referring it to the “Switzerland of the Middle East” with a surge of tourists flowing into its capital, Beirut [39]. Its small surface (10452 Km²) and strategic location on the eastern Mediterranean have allowed tourism to play a leading role in the Lebanese economy with its unique cultural and natural heritage, the landscapes, and climate... [1]. Throughout history, Lebanon has been a crossroads for many different cultures and civilizations [5]. According to Blom Invest Bank [11], tourism is a main source of income and employment contributing to 19% of Lebanon’s GDP in 2017.

Nonetheless, Lebanon’s tourism market revealed significant fluctuations due to internal and external challenges including political instability, low competitiveness, seasonality, and environmental degradation [1]. Since the onset of the Syrian civil war in 2011, Lebanon has entered an economic recession with a 10% drop in its GDP [108]. The Syrian war was considered one of the major humanitarian crises, during which Lebanon accommodated several displaced Syrians. The COVID-19 outbreak coincided with a serious socio-political and economic crisis in the country [48]. Additionally, on August 4, 2020, the Beirut Port’s explosion triggered extensive human and material destruction and had a countrywide economic impact [3]. In 2022, food and beverage costs increased as inflation averaged 171.2% and the currency lost over 98% of its pre-crisis value [108]. Additionally, this sector faces critical environmental challenges causing health concerns for residents and tourists [61]. All these challenges, intensified by weak governance, threaten the sustainability of the tourism industry. A recent regional study assessed that tourism-dependent economies within the Middle East are experiencing increasing sustainability pressures linked to economic resilience, resource scarcity and climate vulnerabilities [74].

According to the UNWTO [120], sustainable tourism involves economic, social, and environmental dimensions, taking into consideration all stakeholders in a specific destination. In this context, to support long-term development, the country must focus its efforts on promoting sustainable tourism as a strategic reform. Implementation of sustainable development in the tourism industry is of particular importance for effective functioning and future success [17, 97, 102]. Research conducted by Gutiérrez et al. [38], emphasized that the hospitality industry is under rising pressure to embrace sustainable practices due to its high customer visibility and noteworthy environmental influence. Sustainability is essential for the lodging industry that uses resources such as water and energy causing significant environmental issues [45, 49, 112]. Recent studies indicate a global trend toward green hotel transformation, where environmental management, green HRM, and energy efficiency are becoming a core aspect of hospitality competitiveness [63].

The topic of sustainability in hospitality has generated a bunch of interest globally with several articles focusing on “sustainable hotels” published between 2003 and 2022. During the last few decades, the bulk of references to corporate sustainability and sustainable strategy have increased extensively [92]. In line with this development, sustainability initiatives have become a central focus for academic writers [117]. In the last ten

years, Harmon [40] highlighted the growing importance of sustainability in the hotel sector. The International Tourism Partnership [46], called for employing the UN SDGs as a central point to guide the responsible business in Hospitality. Berbeka, Alejziak & Berbeka [8], emphasized the need for systematic research on the implementation of SDGs in tourism industry. Yet, while there is an increasing global focus on sustainability research, there are very few scholarly works examining sustainability implementation in the Lebanese hotel industry after the intertwining impacts of an economic collapse, a surge in Syrian refugees, the COVID-19 health crisis, and the Beirut Port explosion [24, 53]. There is some regional work studies focused on either consumer perceptions or tourism development, yet there seems to be a vacuum associated with hotel-level sustainability, restructuring stakeholder commonality, and operationalizing barriers. This research seeks to fill this void by studying how Lebanese hotels enact sustainability practices across the economic, social, and environmental dimensions of the Triple Bottom Line and how stakeholder interactions limit or facilitate these efforts.

The purpose of this paper is to study the sustainability of hotels in Lebanon in its broadest scope. Findings will contribute to identifying valuable strategies that promote sustainable development within the hospitality industry by emphasizing practices that contribute to social, economic, and environmental sustainability among different stakeholders to meet the United Nations' sustainability development goals (SDGs). This research will provide insights for policymakers, hotel managers, and scholars on how to promote effective sustainable development strategies. Due to the limited number of sustainable hotels, this research could help fill existing gaps and improve Lebanon's accommodation services aiming to reach the sustainable development goals. To guide this investigation, the study addresses the following research questions:

- What sustainability practices (social, economic, and environmental) are currently implemented in Lebanese hotels?
- What are the key challenges Lebanese hotels face in implementing sustainable tourism strategies?
- What strategies can promote more effective sustainable development in Lebanon's hotel sector?

The structure of the study is organized as follows: The research context and related literature review along with the theoretical framework are presented, sustainable development in hospitality and tourism, sustainable hospitality practices and challenges are discussed in the subsequent sections. This is followed by the methodology, which covers data collection, analysis, and sampling techniques. The results are then subsequently analyzed. Ultimately, the conclusion delves into the findings, limitations, and future recommendations.

2 Literature review

2.1 Theoretical framework

This research draws on both the Triple Bottom line (TBL) approach along with the Stakeholder approach theory to examine the sustainability practices of hotels in Lebanon. Both theories are used to explore how hotels can effectively implement sustainable practices considering the different interests of stakeholders. Previous research indicates that drawing from both TBL with stakeholder perspectives can help provide insights on

how competing stakeholder interests create tensions in the application of sustainability and how organizations respond to competing interests [99–101].

2.1.1 The triple bottom line approach

The Triple Bottom (TBL) was proposed by Elkington [27]. This approach helps to measure businesses' performance and its success using economic, social, and environmental pillars [35]. This theory provides a holistic framework of the sustainable practices [91]. Economic pillar considers the effect of the organization's business practices on the economic system [25]. It includes the ability of organizations to survive and evolve into the future to support future generations [94]. Social pillar refers to fair business practices to the labor, human capital, and community [25]. It highlights the interaction between the community and the organization and focuses on issues linked to community involvement and employee relations [35]. The environmental pillar engages organizations in clean practices without damaging the environmental resources for future generations [25]. It includes the use of efficient-energy recourses, easing greenhouse gas emissions, and decreasing the ecological footprint, etc [35].

2.1.2 The stakeholder theory

The stakeholder theory was developed by Freeman in 1983 [31]. The theory focuses on the responsibility of an organization to meet the expectations of all stakeholders not only those of the shareholders [29, 30]. As long-term success in response to a problem can be achieved by working thoroughly with all key stakeholders [30]. Stakeholder management facilitates the response of firms to stakeholder argues, thereby setting an influential way of reflecting organizational social responsibilities [28]. In hotels industry, stakeholders include owners, hotel managers, guests and staff who interact in the day-to-day running of hotels [54].

2.1.3 Integrating the triple bottom line approach and the stakeholder theory

Combining both theories will provide a clear framework to analyze the sustainability of hotel practices. The Triple Bottom Line approach assists in evaluating sustainability practices within the hotel sector, ensuring that these practices capture not only financial benefits but also employee well-being, community engagement, and environmental stewardship. Simultaneously, Stakeholder Theory outlines the interactions between hotels and their key internal and external actors, including employees, government, tourists, and the local community—highlighting how decisions and strategies based on effective collaboration affect these stakeholders and frame who should be involved in implementing sustainable strategies. The stakeholder theory lens is an applicable instrument to explore sustainability [54, 114]. The theory provides the tourism industry, including hotels and their management, with a conceptual framework to perform in response the needs and interests of all stakeholders [55]. Combining these two approaches demonstrates that sustainable practices in tourism are affected by the need to balance outcomes across the TBL dimensions and the need to respond to stakeholder expectations and relationships. Peñaflor [84] demonstrates how economic, social, and environmental sustainability practices (TBL) in hotels both influence and are influenced by interactions with key stakeholders, providing a framework to analyze sustainability practices and their impact on customer satisfaction, loyalty, and repeat visits. The study

highlights the strategic value of TBL practices in fostering sustainable hospitality operations and enhancing customer retention, offering actionable insights for budget hotels seeking to align profitability with social and environmental responsibility.

2.2 Sustainable development in hospitality and tourism

In September 2015, the United Nations (UN) launched the Sustainable Development Goals (SDGs) which has become a principal debate area among governments, practitioners, and policymakers internationally [78]. Accordingly, several recent studies focused on the role of the tourism industry in the successful achievement of the SDGs [59, 62, 95].

Sustainable tourism considers three main factors economic, social, and environmental considering all stakeholders in a particular destination [118]. In addition, the participation of all relevant stakeholders guarantees involvement and consensus building of sustainable tourism development [41]. Balancing success with environmental and social responsibilities is a challenge requiring an innovative approach towards strategic management [7, 47, 50, 75, 106]. The significance of maintaining the sustainability of the tourism industry is to maintain tourist satisfaction and warrant a novel experience for them, whilst stimulating their awareness to foster sustainable tourism practices [65]. Therefore, Strategic management in hospitality has developed to adapt to the cumulative demands for sustainability whereas seeking to sustain profitability and competitiveness [41].

However, balancing social, economic, and environmental dimensions in the long-term is one of the biggest challenges for sustainable tourism development [23]. Given the participation of various stakeholders who share different objectives, affiliating their interests towards a common goal becomes an intricate challenge that could be navigated by encouraging stakeholder collaboration and dialogue, tackling probable conflicts, and discovering synergies supporting environmental conservation and economic growth [67]. Henceforth, Sustainable tourism initiatives will play a fundamental role in accomplishing this balance and generating a positive and lasting effect on the environment and the economic prosperity of the communities involved [87].

2.3 Sustainable hospitality practices

2.3.1 Economic sustainability in the hotel industry

Economic sustainability aims to develop balanced and lasting economic growth by promoting profitability, efficiency, and vitality across all levels of society, including the tourism sector [67]. Economic Sustainability focuses on practices that help an organization and its stakeholders realize their requirements without affecting the needs of future [49, 121]. The economic dimension of sustainability is generating profits and increasing a firms' market share while having as few harmful social and environmental effects as possible [98, 121]. According to Parsakia [82], sustainable tourism will contribute to cost efficiency, revenue enhancement and risk management through leveraging green certifications and eco-friendly branding, managing compliance, environmental, and reputation risks and allocating effective budget and employing cost-saving strategies. As Dwyer et al. [23] argued, economic sustainability offers opportunities for the entire stakeholders, counting poverty decrease.

2.3.2 Social sustainability in hospitality

To support sustainable business operations, Green Human Resource Management (GHRM) practices are mandatory to foster an environmentally friendly workplace culture [81]. Additional research highlighted the effect of strategic human resource management (HRM) practices to sustain employees and reduce turnover [93, 125]. The social dimension seeks to advance the welfare of the staff, other suppliers, the visitor, and the surrounding community through proper training, forming a safe workplace, fair trade, building long-term partnerships, and more [36, 45]. Parsakia [82], added that social sustainability in the tourism industry is achieved through stakeholder engagement including involving employees in training programs, incentive schemes, and staff feedback mechanisms, community interaction by hiring locally, supporting community projects, and respecting cultural sensitivities and regulatory bodies through compliance to legal standards and industry guidelines. Sustainability practices also cover a business's responsibilities to society and the environment where many businesses have lately enhanced their operation by adhering to best sustainability initiatives [49, 80].

2.3.3 Environmental sustainability in hotel operations

The subject of sustainability is crucial for hotels as they impact significantly to environmental issues due to the employment of resources such as water and energy [45, 49, 112]. The hotel business is expanding vastly and rapidly making it essential to embrace ecologically sustainable practices [10]. According to Kinyanjui [58], ecological pressures facing hotels in regions are faced through the implementation of water management strategies to minimize the risks of climate change. Dwyer et al. [23] defined environmental sustainability as the preservation of resources focusing on non-renewable ones comprising the carbon footprint, biodiversity, environmental quality, and the capacity to lessen waste, degradation, and exhaustion of resources. The environmental feature of sustainability aims to lessen the adverse environmental effects triggered by hotel operations via proper monitoring mechanisms and, or cooperative efforts like environment-friendly product development and greening of production processes [44, 98]. Parsakia [82], suggested that technological Innovations has revolutionized how to manage and conserve resources.

2.4 Challenges in sustainable implementation

Sustainable tourism development faces economic, social, political, and environmental challenges that affect its implementation [21]. Lack of awareness and education about sustainable tourism, limited stakeholder involvement and absence of adequate funding are among the most common barriers to the implementation of sustainable tourism policies [16, 107]. Additionally, governments, businesses, and local communities often misunderstand or ignore the sustainable tourism policies [15] where many researchers pointed out the lack of strategic planning for implementation of sustainable tourism [76, 88]. Butler [16] suggests that absence of a clear strategic plan for sustainable tourism will result in uncoordinated actions. An additional barrier to sustainable tourism is the excess of bureaucracy [43]. One additional challenge is the balance between the needs of tourists and the local community, preserving resources, and guaranteeing profitability [112].

In Lebanon these challenges are particularly acute due to financial, economic, and health crises resulting from political instability that hinders investment and economic development [64]. Land management practices exacerbate the degradation of landscapes and natural resources who are left without any protection [2]. Moreover, Lebanese attractions remain largely untapped due to ongoing geopolitical conflicts, including political instability, financial and economic turmoil and poor infrastructure that limit tourist inflows and hinder investments in the sector [27]. Additionally, a study conducted by Daou et al. [20], showed that Lebanese Gen-Z have a high level of environmental awareness, but this awareness does not always transfer into actions.

3 Methodology

The current study employed an inductive qualitative methodology to explore the challenges faced by tourism industry in Lebanon and the sustainable practices adopted by hotels in the region to develop stakeholder-driven practices and strategies that promote the sustainable tourism. To offer multiple perspectives on the event under analysis, a multi-case study approach is employed [19, 123]. Therefore, the current study focuses on the perceptions of different stakeholders with different roles working in hotels including general managers, operations and communications managers, HR directors, and engineering supervisors.

3.1 Sampling technique

In qualitative research, purposeful sampling is a widely employed technique for the identification and selection of information-rich cases [83] and the knowledgeable individuals with the phenomenon of interest [18]. In addition to availability and willingness of interviewees to participate, and communicate experiences and opinions in an articulate, expressive, and reflective manner [9, 105].

The current research was conducted with a convenience sample of 15 hotels, from which 4 of them are international chain hotels, 4 boutique hotels, 4 resorts, 3 independent hotels. The sample comprised micro, small, medium-sized, and large hotels locating in different areas of Lebanon spanning from the North through the west and the Beqaa valley, thereby contributing to a broader understanding of the challenges facing hotels of different sizes and their sustainable tourism practices in Lebanon.

3.2 Data collection process

This study surveyed hotels in Lebanon that provide various tourism services (refer to Table 1). A consent form that fully explained the purpose of the research and its dissemination process was prepared and presented to the interviewees prior our interviews to ensure their freely volunteered participation. As robust, clear, and well written consent form is fundamental to avoid weak consent agreement that may compromise the quality of data collected due to mistrust [68]. After obtaining participant's approval, data were collected using semi-structured telephone and in-person interviews. Due to confidentiality principle, the names of participants and hotels are not revealed, and participants are referred to by the codes H1 and H2.

Several studies recommended using a minimum sample size of at least 12 to reach data saturation in qualitative research [13, 32, 37]. Therefore, a sample of 15 hotels was deemed sufficient for the qualitative analysis and scale of the study. Moreover, no new

Table 1 Study participants and their demographic distribution

Participant	Position	Gender	Hotel category	Location
H1	Operations manager	Male	Boutique hotel	Qalmoun, North Lebanon
H2	HR department	Female	International chain hotel	Dbayeh, Mount Lebanon
H3	Engineering supervisor	Male	International chain hotel	Baalbak, Bekaa
H4	HR	Male	International chain hotel	Beirut
H5	Manager	Female	Boutique hotel	Al Mina, North Lebanon
H6	Manager	Male	Resort	Qammouaa, Akkar
H7	HR	Female	Boutique hotel	Al Mina, North Lebanon
H8	HR	Female	Resort	Al Koura, North Lebanon
H9	Operations manager	Male	Independent luxury hotel	Tripoli, North Lebanon
H10	Operations manager	Female	Medium-sized, modern boutique hotel	Beirut
H11	Communications manager	Male	Resort	Tripoli, North Lebanon
H12	Communications manager	Female	Independent small hotel	Mount Lebanon
H13	HR	Female	Independent small hotel	Tripoli, North Lebanon
H14	Communications manager	Female	Eco-friendly resort	Kaftoun, North Lebanon
H15	Quality control manager	Male	Resort	Chouf, Mount Lebanon

themes emerged after the final five interviews, confirming sufficient coverage of the topic. The hotels were selected for the study based on their accessibility and willingness to share their perspectives and practices.

3.3 Data analysis: a thematic analysis approach

To interpret the primary data collected for this research, thematic analysis was employed following the methodological approach advocated by Braun and Clark [13] and adopted in similar studies in Saudi context, such as Elgammal et al. [26]. This method interprets and classifies existing themes to refine the qualitative data [12] using NVivo software 12. Direct quotations from interviewees were used to code and categorize the data according to similarities and differences [68]. As a result, numerous themes were generated to interpret the study's results [14]. All data were audio-recorded, transcribed, and verified by participants to ensure accuracy and reliability [69].

- **Initial Thematic Framework:** Before analyzing the data and based on the study's theoretical foundation and relevant literature review, a preliminary list of general themes was developed to guide the focus on key dimensions of the research question.
- **Inductive Thematic Analysis:** Using NVivo12 software, an inductive coding process was undertaken. The data were carefully examined to identify recurring ideas, perceptions and experiences shared by participants, from which codes were generated. Direct quotations were used to support the themes.
- **Iterative Refinement:** Multiple rounds of coding and analysis were performed to refine the themes, merge overlapping codes and separate nuanced patterns into subthemes. Due to resource limitations, coding was conducted by a single researcher with regular consultation with the research team to reduce biases and ensure interpretative accuracy.
- **Linking Themes to the Theoretical Framework:** The final set of themes was mapped back to the Triple Bottom Line (TBL) and Stakeholder frameworks to interpret how the findings aligned with or extended existing theoretical perspectives.

4 Findings and results

4.1 The current state of tourism and development in Lebanon

4.1.1 Favorable and detrimental elements affecting tourism demand

As presented in Fig. 1, participants highlighted several positive factors that affect tourism demand in Lebanon including the stunning nature and weather (H4, H6, H12, H15), as well as the warm hospitality, authenticity, and the diverse cultural, historical, and natural attractions (H2, H7, H9, H10, H12, H15). H11 stated: *“we really know how to make people feel welcome. The Lebanese natural landscapes, from the mountains to the coast, are a huge attraction for those wanting to relax and enjoy nature”*. Others also pointed to the vibrant nightlife and festivals and many entertainment activities (H3, 10). H12 said: *“People come here for hiking, skiing, and enjoying the peace and quiet”*. One of the hotels also highlighted the renowned Lebanese cuisine (H9).

In parallel, interviewees mentioned multiple negative factors threatening the tourism industry in Lebanon. Most participants (H1, H5, H9, H10, H12, H13, H14, H15) revealed the economic crisis, which they linked to several challenges such as the Syrian refugee influx, the Covid-19 pandemic, Beirut explosion, and Lira devaluation. One participant (H7) highlighted the rising costs of goods and services, making sustainable operations difficult. Other participants (H1, H2, H5, H9, H11, H14, H15) noted the political instability, which made many travelers hesitant to visit the country. Most of the hotels (H1, H2, H3, H4, H10, H13, H14) declared that the recent war between Israel and Lebanon generated security concerns regarding tourist safety. Some hotels showed concerns about the environmental issues arising from pollution, waste, and poor resource management (H8, H11). Other hoteliers (H1, H6, H8, H12, H14, H15) mentioned limited infrastructure and a lack of Government support. H13 confirmed: *“the overall image of Lebanon in the international news can sometimes be negative due to regional issues and the limited development initiatives”*. Others (H6, H13) cited the lack of marketing, to attract tourists, which has led to a decrease in demand for tourism in Lebanon (H6, H13) (Fig. 1).

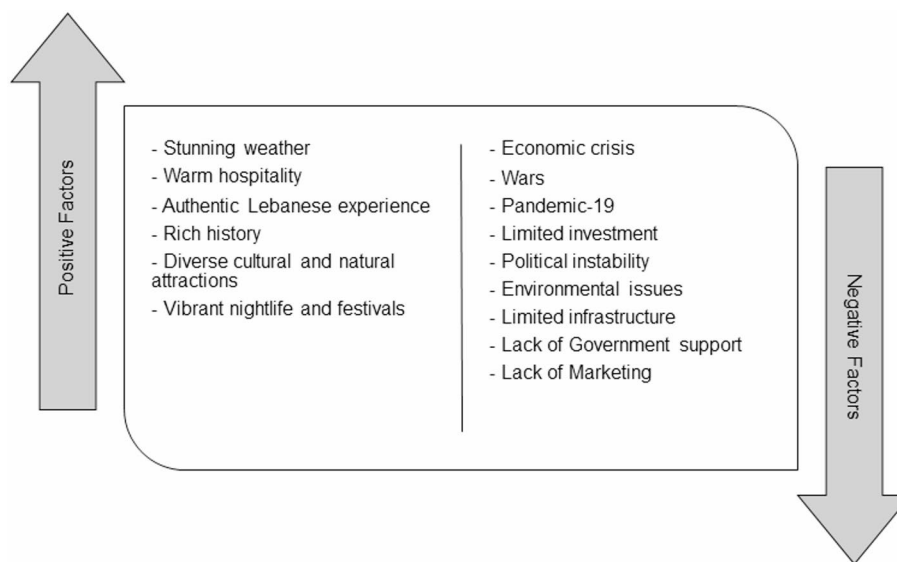


Fig. 1 Positive versus Negative factors affecting the tourism demand in Lebanon including key drivers such as weather, hospitality, authenticity, history, natural and cultural attractions, and barriers such as political instability, infrastructure deficiencies, crises, weak government based on insights from stakeholder interviews

4.1.2 Hotels perceptions on the concept of sustainability in the tourism industry

As illustrated in Fig. 2, hotels (H6, H10, H12, H14) perceived the concept of sustainability as being conscious of the environmental impact and trying to minimize its effects on the surrounding ecosystem. Hotel H9 considers the “green initiatives” as an underlying principle of their operations. One participant (H8) reported that sustainability is about being mindful of water consumption through supporting water-saving systems and responsible marine activities. H11 confirmed by stating: *“we focus on environmental conservation efforts, such as beach cleanups and working with local organizations on marine preservation.”* Other hotels focused on energy conservation such as energy-efficient lighting (H5, H9, H10, H14) while some emphasized the importance of waste management (H6, H9, H12, h14). Other Hotels (H2, H7, H10, H13, H14) defined social sustainability as treating tourists like family, meeting their needs, and providing quality services. Hotels H12 and H13 added that sustainability is about encouraging guests to be mindful of their environmental footprint. Several hotels (H9, H12, H13) perceived sustainability as supporting the local community through sourcing local products and employing local staff. While one hotel (H19) reported promoting local and eco-friendly amenities. H11 established: *“We try to work with local fishermen and businesses to ensure our presence benefits the whole community.”* Hotels H7 and H9 defined sustainability as protecting cultural identity and preserving the beauty of the hotel’s surroundings and Lebanese heritage for future generations. However, only Hotel H2 highlighted the importance of meeting the needs of employees through training and hiring the best from hospitality graduates. In parallel, H15 mentioned: *“Few hotels are becoming aware of sustainable tourism, but not all apply it seriously. Some follow it because it’s a trend”* (Fig. 2).

4.2 Practices implemented to promote sustainable tourism in hotels

4.2.1 Environmental sustainability of the hotel sector

Participants reported that hotels in Lebanon have employed several initiatives to promote sustainable tourism highlighting waste management, water conservation, and energy consumption reduction. Some hotels (H5, H9, H10, H11) invested

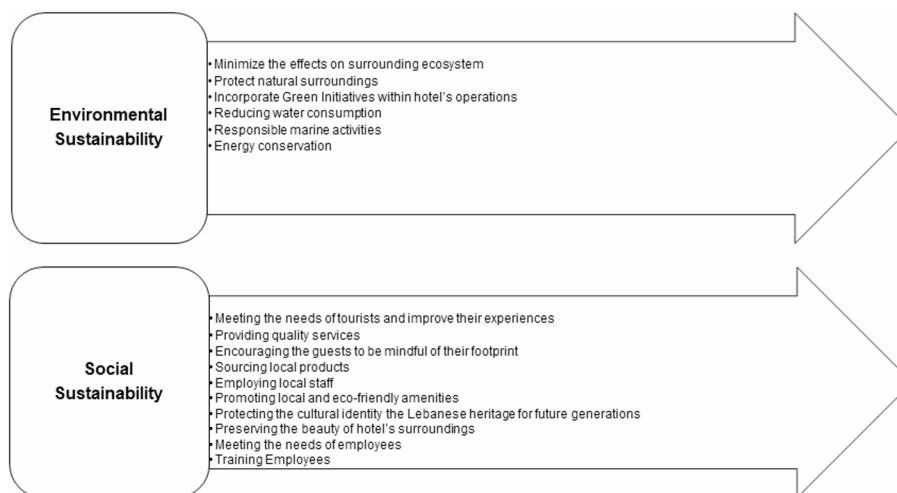


Fig. 2 Participants' conceptualizations of sustainability in the tourism industry including key interpretations such as environmental protection and social responsibility based on data collected from authors' interviews

in water-efficient fixtures in guest rooms and public areas such as, using water-saving devices, water-saving showerheads, and faucets, with regular monitoring of their water consumption. Other hotels (8, H15, H11, H15) treat greywater through efficient irrigation systems. Hotels H6 and H12 conserve water using rainwater harvesting systems and water-efficient fixtures, while hotels (H1, H2, H11, H12) emphasized their awareness programs on water consumption among their employees and guests.

Regarding energy reduction, most participants (H2, H5, H9, H11, H12, H13, H15) revealed gradual upgrading and replacement of older appliances with energy-efficient options, such as using LED lighting. Other hotels added the installation of solar water heaters (H8, H12, H13) and management systems to optimize energy consumption based on occupancy and time of the day (H9, h10). Most of the hotels (H1, H6, H11, H13, H14, H15) also invested in solar panels alongside generators.

Most of the participants (H5, H4, H9, H11, H12, H13, H15) reported managing waste reduction through programs including sorting recyclables, composting organic materials, and reducing overall waste sent to landfills. H10 confirmed the above and added: *“We placed recycling bins throughout the hotel, and we’re working towards reducing single-use plastics in our amenities and food service. We also partner with local recycling initiatives to ensure proper disposal and processing of our waste.”* Hotel H1 and H3 mentioned a lack of application of sustainability practices regarding waste management, as illustrated in Fig. 3.

4.2.2 Economic sustainability of the hotel sector

Figure 4 illustrates that, to support economic sustainability, several participants (H2, H5, H8, H13, H15) focused on efforts to increase the number of guests by maintaining high services standards and offering premium eco-packages aimed at attracting eco-conscious guests willing to pay more. Furthermore, some hotels (H14, H10, H15) reduced their long-term operational costs through investment in eco-friendly infrastructure. H9 established: *“We invest in energy-efficient technologies, even with higher costs, but they ultimately enhance the hotel’s reputation and attract discerning travelers who value authenticity and responsible practices.”* Other hotels (H1, H6, H9, H13, H14, H15) reduced their costs by using locally sourced food and handmade furniture. Hotels H1 and H15 increased the employment rate by hiring local staff. Only two hotels (H10, H12) used marketing initiatives to promote their brand identity as eco-friendly hotels,

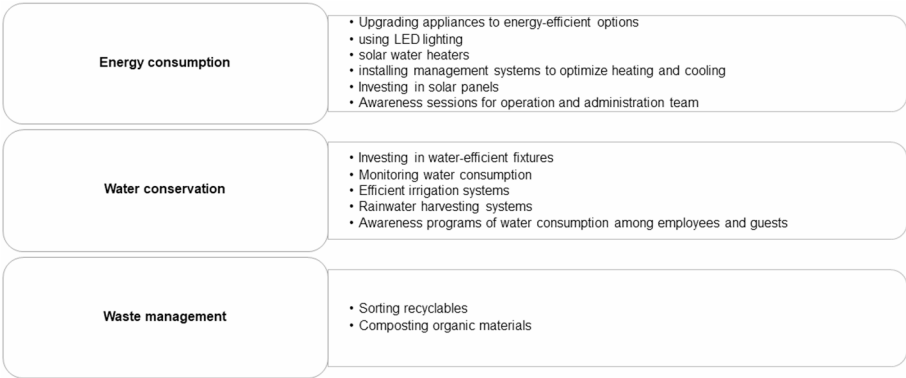


Fig. 3 Practices implemented to promote environmental sustainability of the hotel sector including reduction of energy consumption, water conservation and waste management based on data collected from participant interviews

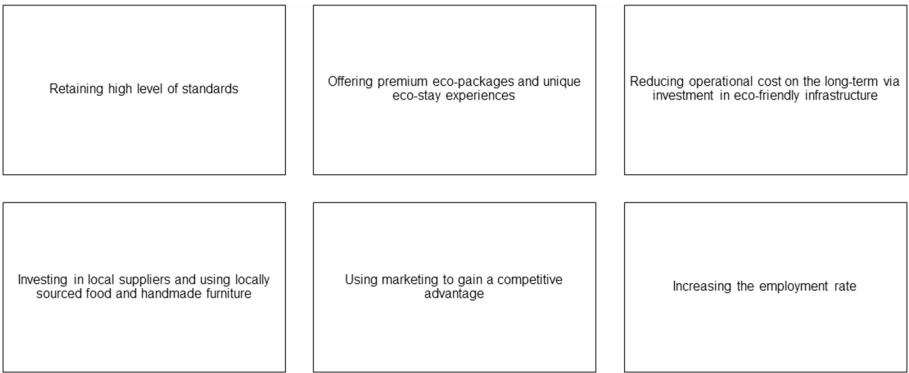


Fig. 4 Practices implemented to promote economic sustainability of the hotel sector including cost-cutting measures, diversification of services, maintaining quality, local sourcing, and staff retention based on data collected from participant interviews

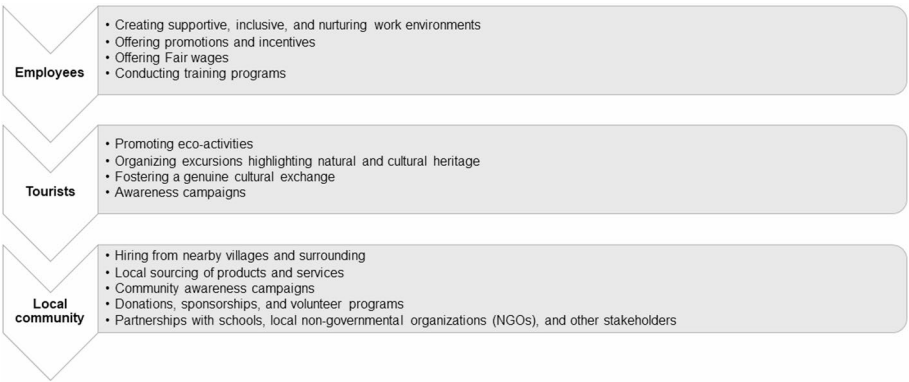


Fig. 5 Practices reported by hotel stakeholders and implemented to promote social sustainability of the hotel sector reflecting the hotel's commitment to social responsibility toward employees, tourists, and the local community

aiming to gain a competitive advantage and attract guests willing to pay a premium for a stylish and responsible stay (Fig. 4).

4.2.3 Social sustainability of the hotel sector

As depicted in Fig. 5, to support social sustainability, most interviewees conducted regular training programs and development opportunities on sustainability, cultural sensitivity, and responsible tourism practices to enhance staff skills (H2, H4, H5, H9, H10, H11, H12, H14, H15). Some hotels (H1, H10 and H13) mentioned the creation of a supportive, inclusive, and nurturing work environment. Other hotels (H1, H10, H15) offered promotions and incentives to satisfy employees in addition to paying fair wages.

For the tourists, hotels (H5, H11 and H12) highlighted eco-activities that emphasize the region's natural and cultural heritage, creating a positive impact on the community and the environment while fostering genuine cultural exchange. H9 confirmed by stating: *"We offer cultural immersion experiences, such as guided tours of historical sites and traditional Lebanese cooking classes, fostering a deeper understanding and appreciation of our culture."* Most participants (H6, H8, H10, H10, H11, H12, H13, H14, H15) also focused on awareness campaigns and guest education about the local culture and responsible tourism practices.

Regarding the local community, most participants (H1, H6, H9, H11, H13, H14, H15) revealed creating job opportunities through local hiring. Many hotels (H5, H8, H9, H10, H14, H15) mentioned supporting the local economy through partnerships with artists, artisans, and farmers to source products and services. H12 declared: *“We actively collaborate with local farmers and producers, offering our guests fresh, locally sourced ingredients and supporting sustainable agriculture.”* Hotels H3 and H7 conduct community awareness campaigns, while Hotels H2 and H11 focused on social responsibility initiatives, supporting local schools and community through donations, sponsorships, and volunteer programs. Hotels (H7, H9, H14) reported partnerships with schools, local non-governmental organizations (NGOs), and other stakeholders to support community-based tourism development projects. Only one hotel H1 admitted that: *“We still lack deep information about sustainability in relation to humans”* (Fig. 5).

4.3 Challenges of sustainable tourism facing hotels in Lebanon

According to Fig. 6, participants revealed several challenges hindering sustainable tourism initiatives. Environmentally, most participants (H1, H5, H6, H7, H9, H10, H12, H13, H14, H15) reported budget limitations, especially in the current economic situation, as eco-friendly alternatives can be more expensive. Hotels H9, H10 and H12 discussed the lack of infrastructure development. Interviewees focused also on the crises represented by the recent wars and the unstable political situation in the region (H1, H2, H3, H4) in addition to climate change (H11).

Economically, H10 and H14 admitted that maintaining affordability while being eco-conscious is major challenge. Hotels H1 and H2 revealed that low occupancy can lead to reduced revenues and job losses, which slow economic growth.

Socially, hotels highlighted the lack of awareness among both guests and staff to promote sustainable practices, stating that changing habits is challenging due to resistance to change (H1, H5, H6, H9, H11, H12, H14, H15). Hotels H4 and H7 faced limited technical knowledge among staff, while hotel H2 emphasized high staff turnover. Limited government support and lack of collaboration with local authorities were among the challenges cited by hotels who mentioned the absence of plans to manage natural resources and inconsistent policies (H1, H6, H7, H8, H10, H11, H14) (Fig. 6).

Environmentally	Socially	Economically
• Budget limitations and lack of securing funding • Lack of infrastructure development • Crises and disasters • Climate Change	• Lack of awareness among both guests and staff • Resistance to change • Limited technical knowledge • High turnover • Limited government support • Lack of collaboration with local authorities • Absence of plans to manage natural resources sustainably and conservation efforts • Inconsistent policies	• Balancing between cost of sustainable products and competitive pricing • Low occupancy

Fig. 6 Challenges impeding the implementation of sustainable tourism practices in Lebanese hotels including environmental challenges such as limited financial resources, inadequate infrastructure, natural disasters, and climate change, social challenges such as lack of awareness, government support, collaboration, limited knowledge, and resistance to change, and economic challenges including maintaining quality with competitive prices based on data collected from interview participants

5 Discussion and implications

Lebanese tourism is considered the backbone of the country's economy, generating most of its revenue streams. Bassil et al. [6] stated that tourism makes a significant contribution to Lebanon's gross domestic product (GDP). It plays a leading role in the Lebanese economy due to its cultural and natural heritage, varied landscapes, pleasant climate, and strategic location [1]. In this regard, participants praised the opportunities that this hospitable country offers, such as its favorable climate and rich history, culture, and natural attractions. However, the current state of tourism in Lebanon is impacted by a numerous unfavorable factors that affect demand, such as repeated wars, political unrest, inadequate infrastructure, and a lack of government support, all of which have contributed to the collapse of the industry, a drop in state revenues, and reduced investments, making many tourists reluctant to travel to the country, thereby limiting development opportunities. According to a report by Kataya [52], the Covid-19 pandemic, the August 2020 Beirut Port explosion, and the severe economic recession all contributed to a decline in tourism in 2020 compared to 2019, in addition to political instability, low competitiveness, seasonality, and environmental degradation [1].

According to Omair et al. [78], sustainable strategies in the tourism industry are suggested to manage the economic, social, and environmental impacts while addressing the needs of various stakeholders. Henceforth, there is growing attention on promoting sustainable tourism practices to revive this sector in Lebanon, considering the economic, social, and environmental impacts, considering the different needs of stakeholders and contributing to the achievement of the Sustainable Development Goals. Sustainable tourism contributes to all SDGs across economic, social, and environmental domains, either directly or indirectly [116]. Nevertheless, when asked about participants' understanding of the concept of sustainability, hotels primarily focused on the environmental and social pillars of sustainability. Although hotels are becoming more aware of sustainable tourism, there is a lack of practical application, as seen in the economic pillar being largely overlooked. This confirms the study conducted by Budeanu et al. [15], which suggested that sustainable tourism is often misunderstood or ignored by the main players involved in tourism development, such as governments, businesses, and local communities.

Practically, hotels in Lebanon are fostering environmental sustainability by first investing in water management and conducting awareness campaigns for staff and visitors that encourage the prudent use of energy and water. Second, hotels are progressively upgrading their appliances to energy-efficient options, using LED lighting, solar water heaters, and solar panels alongside generators to reduce energy consumption. Third, hotels adopt waste management systems that include composting organic materials and sorting recyclables. Olya et al. [77] supported participants' practices by explaining environmental sustainability as the process of taking proactive measures through the implementation of appropriate environmental systems and technologies. Nagendrakumar et al. [73] added that the hotel industry is "going green" to reduce its environmental effect by using sustainable suppliers and waste-reduction strategies. Other studies provided examples of environmental practices employed by hoteliers, including the careful disposal of solid waste, the use of environmentally friendly cleaning products, and energy- and water-saving methods [4, 70].

Economically, Hotels reduce operating costs by investing in energy-efficient technologies and eco-friendly infrastructure, hiring local suppliers, and using locally sourced products and services to balance economic growth with sustainable tourism activities. Several researchers agreed, stating that economic sustainability represents a win-win scenario created by hoteliers through the implementation of effective sustainable practices that benefit both the environment and generate significant long-term profits [51, 57, 86]. Nagendrakumar et al. [73] and Pereira et al. [85] also confirmed, defining economic sustainability as encompassing organizational attributes that must be maintained over time. Olya et al. [77] and Kularatne et al. [60] noted that implementing green practices is not only a moral responsibility but also offers advantages in terms of cost savings, enhanced reputation, improved market positioning, stronger social commitment, guest retain, and increased revenue. Hotels further support economic sustainability by hiring local staff, offering premium eco-packages, and providing distinctive eco-stay experiences that appeal to tourists willing to pay more for high value. This aligns with a study suggesting that eco-friendly branding attracts a niche market segment that values sustainability [82].

Socially, Hotels create a supportive and inclusive work environment, fair wages and incentives, local employment, and training programs for employees. Hence, in the Lebanese context, Hotels demonstrated a strong commitment to social sustainability of their employees. However, the situation in Marsa Alam's tourism sector is quite different. According to Mekawy [66], in Marsa Alam's tourism sector, business leaders acknowledged a lack of social sustainability regarding their employees, noting the absence of a safe workplace and employee well-being, as well as the need for long-term social security policies to protect workers' rights and sustain resilience, especially during crises. It can be concluded that Lebanon is in a relatively safe position in terms of the social sustainability of its employees. Regarding the tourists, hotels place a strong emphasis on protecting the environment and cultural heritage by educating visitors about local customs and traditions and encouraging them to support local businesses. This is comparable to other research that examined how the social component of sustainability aims to improve the well-being of employees, vendors, visitors, and the local community overall by providing appropriate training, creating a safe work environment, practicing fair trade, forming enduring relationships, and more [36, 45, 96, 111]. For the local community, hotels hire and partner with locals to source products and services, creating job opportunities and involving the community in the tourism industry. Furthermore, hotels collaborate with schools, non-governmental organizations (NGOs), and other stakeholders to support community-based tourism development projects, contributing to the achievement of SDG17, which focuses on partnerships among different stakeholders. Some hotels promote social responsibility through donations, sponsorships, and volunteer programs to reduce poverty, support education, and sustain cultural festivals. This aligns with the findings of Parsakia [82], who demonstrated that social sustainability fosters community interaction through engaging with the local community via employment practices, supporting community developments, and safeguarding cultural sensitivity.

Therefore, social, economic, and environmental sustainability are strongly related [53], leading to a stronger case for the SD [78]. Initiatives to improve environmental sustainability in Lebanese hotels demonstrate alignment with SDG 12 and 13. As they incorporated responsible operational practices that highlights the sustainable management and

efficient use of natural resources. Economically, hotels proved progress toward SDG8 by promoting sustainable tourism which supports the local economy and employment growth. Social sustainability practices adopted by Lebanese hotels contributed directly to the achievement of SDG 8 that aim for decent work and economic growth, SDG4 that promote quality education and SDG 11 that supports sustainable cities and communities. This is consistent with the studies of UNWTO and UNDP [120] that emphasized the role of tourism in the contribution of SDGs and mitigating the sector's negative impacts through strong collaboration of all tourism stakeholders. Hence, tourism play a key role in reaching “decent work and economic growth” (SDG 8) and “responsible consumption and production” (SDG 12) [89, 120]. Similarly, a study in South Asia examined the significance of tourism for the economic development of the country as well as the achievement of SDGs such as poverty reducing poverty [108].

Although hotels in Lebanon are moving toward more sustainable tourism practices, some challenges hinder their ability to achieve successful sustainability including, limited access to funding, inadequate infrastructure development, crises and disasters, an unstable political situation, and climate change. Those challenges are supported by Onn et al. [79] revealing that sustainable practices are hampered by technical difficulties such as limited infrastructure, insignificant grid connectivity, and a shortage of renewable energy options. Wang et al. [115] added, promoting sustainable energy techniques can be expensive initially making it difficult for hotels to make the investment pay off. Dias, Viana & Pereira [21] confirmed by stating that the lack of financial incentives can push private sector to refrain from investing in sustainable tourism. The primary economic difficulty is striking a balance between competitive prices and cost of sustainable products, as well as the low occupancy rate resulting from geopolitical environment that slows economic growth. A researcher affirmed that balancing between the needs of guests and the local community, conserving resources, and generating profits at the same time are among the main challenges [113]. Socially, hotels reported resistance to habit change among employees and guests due to limited awareness of sustainable practices, as well as challenges posed by a lack of skilled workforce and high staff turnover. This affiliates with several studies showing that governments, businesses, and local communities are often unaware about sustainable tourism and may misunderstand or ignore the sustainable tourism policies [15, 16]. Finally, hotels highlighted a lack of community engagement and limited support from government and local authorities, including insufficient plans, policies, and conservation efforts to promote sustainable tourism. This aligns with the study of Yacoub [122], who proposed that legal constraints are one of the main obstacles' hotels have while employing sustainable energy measures accentuating the importance of regulatory agencies and hoteliers acting together to foster advantageous regulations that advance sustainable energy practices. Mekawy [66] confirmed that, to achieve the triple bottom line approach, government funding and support are required, along with stronger private-sector participation in decision-making to sustain the tourism industry.

5.1 Theoretical implications

The study contributes to the theoretical development of the Triple bottom Line approach [27] and the Stakeholder theory [31], which together offer a holistic lens for analyzing sustainable practices in the hotel industry. The results exhibit that hotels in Lebanon

Table 2 Recommended sustainability strategies by hotel category

Recommended sustainable practices	Hotel category			
	Independent hotel	International chain hotel	Boutique hotel	Resort
Transparent reporting of performance	Possible but may need external support	Recommended due to the availability of resources	Possible with tailored and simpler reporting	Highly recommended due to stakeholder demands
Enforcing green HR policies	Possible with resource limitation	Recommended due to strong HR structure	Possible especially for personalized trainings	Vital due to large staff
Performing environmental audits	May need external help	Recommended	Possible	Recommended
Seeking partnerships	Recommended access resources	Partnerships with corporate & NGO	Recommended	Recommended
Pursuing green certifications	Limited to resources availability	Crucial for brand reputation	Recommended to appeal to clientele	Important to attract eco-conscious guests
Fostering green procurement policies	Recommended	Recommended	Recommended by focusing on local/organic suppliers	Recommended
Fostering innovation and technology	Limited to high cost	Recommended due to the availability of resources	Recommendation for differentiation	Important to enhance guest experience

are engaging in environmental, economic, and social sustainability initiatives, supporting the TBL framework. However, given the resources scarcity that hotels confront, the economic crisis in Lebanon can often result in putting economic survival ahead of social or long-term environmental sustainability. Similarly, weak governance resulting from political instability affects the practical application of TBL theory due to limited regulatory enforcement and lack of monitoring. Henceforth, sustainable tourism in the hotel sector is not limited by the role of hoteliers only but by a collective effort engaging all stakeholders. Therefore, the findings emphasized the important role of different stakeholders in achieving sustainable tourism focusing on employees, tourists, local community, government, and local authorities. Although, the stakeholder theory assumes equal involvement of all stakeholders, the Lebanese context, where employees are underpaid due to the financial crisis and regulations are not enforced due to weak governance, creates disproportionate responsibilities, especially with the absence of the government allowing powerful stakeholders like hoteliers and investors to dominate the sustainability agenda. This suggests that the interconnectedness of both the Triple bottom Line (TBL) and Stakeholder theory must be reconceptualized in a crisis setting where there are weak state institutions. The Lebanese setting demonstrates that hotels cannot be expected to meet the three pillars of sustainability without stakeholder collaboration that becomes essential with the limited government involvement to turn sustainability goals into action.

5.2 Practical implications

Some recommendations are provided to the hotel industry, as well as to other stakeholders, such as the government and local authorities, to effectively address the challenges that hinder sustainability efforts in the tourism sector and to promote overall sustainable development (Table 2).

5.2.1 For hotel industry practice

Transparent reporting of performance Creating a committee to exchange guest and staff practices, loyalty and experiences this will enhance transparency through clear labeling and public reports that inform visitors and the community at large about sustainable initiatives and advancements. However, it is difficult to measure and report company's performance to the public considering the three measures of sustainability [124].

Enforcing green HR policies Establishing sustainability task forces, encouraging thorough training on sustainable tourism practices, and implementing green recruitment practices. Palguna [81] focused on the crucial role of Green Human Resource Management (GHRM) in promoting an environmentally friendly workplace culture that fosters sustainable business processes.

Performing environmental audits Identifying key areas for improvement and setting clear, measurable targets. Research conducted by Parsakia [82], shows that regulatory compliance is essential to achieve sustainability through obedience to legal standards and industry procedures, supported by rigorous compliance checks and regular audits.

Seeking partnerships Partnering with local and international environmental organizations. Enhancing sustainable efforts is based on encouraging strong relationships with local communities and regulatory bodies [82].

Pursuing green credentials Such as eco-certification, industry-wide sustainability certifications, and ISO14000. Evidence suggests that boosting sustainable tourism depends on revenue enhancement that is based on the leverage of green certifications and eco-friendly branding [82].

Fostering green procurement policy Implementing a "green procurement" policy that prioritizes environmentally friendly products and services. Sustainability is contingent upon process improvements through the optimization of the workflow, using eco-friendly logistics, and green procurement processes [82].

Benefiting from Innovation and technology through:

- *Leveraging AI-powered systems:* installing smart building management systems and employing apps for energy monitoring to automate and regulate heating, cooling, and lighting to maximize energy use based on occupancy and current requirements.
- *Digitalization of services:* Implementing digital solutions for guest's services such as digital menus and online check-in/check-out, as well as a basic online booking and communication system.
- *Digital Marketing and ethical marketing initiatives:* using online platforms to promote responsible travel green offers and local sustainable activities to our guests, encouraging them to engage with the community responsibly.

Literature reveals that managers should focus on leveraging technology to enhance resource management. As pointed out by Parsakia [82], sustainability depends on adaptation of technological innovations covering the employment of IoT for energy management, automated waste sorting systems, and advanced water reclamation technologies. Additionally, digital platforms play a crucial role in broadcasting information and establishing visibility via digital storytelling and interactive content [33] (Table 2).

5.2.2 For policy makers

Short-term Strategies

- *Training and Awareness campaigns:* fostering educational programs and awareness campaigns for both the industry and tourists on the importance of sustainable tourism. Dwyer et al. [22] proposed that Understanding the implications of sustainable development and the guiding principles for promoting sustainable tourism are essential components of tourism planning. Communities should therefore be well-informed about the tourism industry, its impacts, and the processes for planning, fostering harmony, and resolving disputes between all stakeholders involved.
- *Targeted marketing campaigns:* promoting Lebanon as a sustainable travel destination through national and international campaigns that highlight the genuine experiences that showcase the nation's cultural and natural diversity, which can help smaller, community-based accommodations. The development of sustainable tourism is a vital strategic marketing approach to the management of a WH site [34].
- *Implementing green certification programs and offering incentives and grants:* Policymakers can incentivize sustainable initiatives by implementing a green certification program for tourism businesses who support eco-friendly projects or offer incentives alike decreasing taxation and fees or subsidies on hotels who are implementing eco-friendly tourism activities, conserve biodiversity and empower local communities. This coincides with the study of Sobaih et al. [103] who suggest that governments and policymakers should motivate small lodging enterprises to enhance their green ability and seek green innovation to achieve positive environmental performance. Additionally, Germany's government is designated by robust eco-certification programs, with latest studies examining their effectiveness in promoting measurable sustainability advancements within the hospitality sector [72]. While Germany provides a good example of how to implement strict environmental regulations, it is important to note that its strong institutional and governmental support differs from Lebanon's governance context, which requires more assistance as recommended by the long-term strategies.

Long-term Strategies

- *Enforcing clear environmental standards and regulations:* to encourage eco-friendly practices in the travel industry, provide a framework for hotels and other travel operators to adhere to, and establishing clear guidelines and national sustainability standards for putting sustainable practices and technologies into practice. The main importance of proactive political leadership is witnessed in the persistent commitment to developing green requirements and comprehensive sustainability legislation [71].
- *Establishing collaboration platforms:* providing platforms for tourism businesses where users can access a wide variety of resources, share their own practices, their achievements, and projects, induce discussion and cooperation that is deliberated to advance the accomplishment of the sustainable development goals via tourism.
- *Investing in sustainable infrastructure:* such as efficient waste management systems and renewable energy sources and promoting eco-friendly public transportation

options for tourists. Recent research has highlighted the requirement for sustainable infrastructure and policy frameworks [104].

- *Effective governance and preservation efforts*: preventing pollution by establishing protected marine areas and conservation of marine ecosystems to protect natural resources. Inconsistencies between government structures and perspectives lead to the dominance of local and private interests over socially fair development that considers the needs of local community [90].

6 Limitations and implications for future research

While the study offers valuable insights into sustainable practices and challenges in Lebanese hotels, several limitations should be acknowledged. First, our sample is limited in diversity where most of hotels were in North Lebanon, with 11 local hotels, whereas only four were international hotels. Consequently, the sample may not fully reflect the views of all types of hotels in Lebanon, especially international ones which may affect the applicability of the findings to globally hotels. Future research should involve more international chain hotels with a comparative study across different countries around the world to cover diverse perspectives applicable to the global hotels sector. Second, the study relied solely on qualitative interviews as a data collection tool which may limit the depth of our findings. Third, semi-structured interviews were employed to collect data, therefore there might be potential for interview bias as participants may have presented responses that are socially acceptable or that meets their personal or professional interests, which could affect the objectivity of the findings. Future research could strengthen these findings by incorporating mixed methods or quantitative data to triangulate results using.

7 Conclusion

Reaching sustainable development goals mentioned by the UN is achieved through the collaboration between different sectors. Tourism is one of those sectors where sustainability efforts require coordination among different stakeholders successfully contributing to the achievement of the SDGs. Lebanon is considered a hospitable country and a destination for many tourists worldwide. However, weak governance, limited infrastructure and political instability limit the development of this sector. Hence, Sustainable tourism is an ideal way for hotels in Lebanon to recover and develop a sustainable tourism future.

Although hotels in Lebanon are moving toward more sustainable practices in their operations in terms of environmental, social, and economic domains, the sector faces several challenges that impede their sustainability efforts. Therefore, based on the Lebanese context, further suggestions are made to assist hotels in promoting sustainable tourism and effectively accomplishing SDGs, considering the three pillars of the triple bottom line approach and the participation of all stakeholders. As for the hoteliers, sustainable practices should be supported by transparent reporting of performance, green HR policies, environmental audits, Ethical marketing, collaboration among environmental organizations, green certifications, green procurement policy, interactive programs to engage tourists and Innovation and technology. In parallel, policymakers should provide incentives and grants that support sustainable projects, marketing campaigns,

collaboration platforms, green certification programs, clear environmental standards and regulations, investment in sustainable infrastructure, training and awareness campaigns and conservation efforts.

In conclusion, effective collaboration among stakeholders, including hotel employees, shareholders, visitors, the government, and local communities is essential to drive sustainable tourism practices and achieve the SDGs.

Acknowledgements

Special recognition goes out to all interview participants who helped to accomplish this research.

Author contributions

All authors contributed equally to this work.

Funding

The authors did not receive support from any organization for the submitted work.

Data availability

The datasets produced during the current study are available from the corresponding author upon request.

Declarations

Ethical approval and consent to participate

Approval was obtained from the ethics committee of City University. The procedures employed in this study follow the tenets of the Declaration of Helsinki.

Consent for publication

All participants consented to the publication of anonymized data and findings derived from this study.

Informed consent

Consent was obtained from all participants to take part in this study.

Competing interests

The authors declare no competing interests.

Received: 23 May 2025 / Accepted: 8 December 2025

Published online: 21 December 2025

References

1. Abou Arrage J, Hady SA. Ecotourism and sustainability: practices of the Lebanese nature-based operators. *Menadzment U Hotelijerstvu I Turizmu*. 2019;7(1):11–23.
2. Abou Arrage J, Khreis A, El-Kurdi A, Mikhael M, Nasr F. Heritage and landscape sustainable management in the tourism industry. In: *Leadership and governance for sustainable tourism: proceedings of summer school*. 2014. pp. 53.
3. Al Jazeera. Lebanon crisis timeline: from Beirut blast to gov't's resignation [Internet]. 2020 [cited 2025 May 11]. Available from: <https://www.aljazeera.com/news/2020/08/lebanon-crisis-timeline-beirut-blast-gov-resignation-200810171817351.html>
4. Alonso-Almeida M, Llach J, Marimon F. A closer look at the 'Global reporting initiative' sustainability reporting as a tool to implement environmental and social policies: A worldwide sector analysis. *Corp Soc Responsib Environ Manag*. 2014;21:318–35.
5. Baba I. Ecotourism: a factor of sustainable development? Case of North Lebanon [dissertation]. Perpignan: University of Perpignan; 2013.
6. Bassil C, Saleh AS, Anwar S. Is tourism in Lebanon subject to permanent or transitory exogenous shocks? *Tourism Anal*. 2014;19(6):781–90.
7. Bavik A. A systematic review of the servant leadership literature in management and hospitality. *Int J Contemp Hosp Manag*. 2020. <https://doi.org/10.1108/ijchm-10-2018-0788>.
8. Berbeko K, Alejziak W, Berbeko J. Sustainable development goals of agenda 2030 in the declarations and aims of international tourism organisations. *J Travel Tour Mark*. 2024;41(1):142–53.
9. Bernard HR. *Research methods in anthropology: qualitative and quantitative approaches*. 3rd ed. Walnut Creek (CA): Alta Mira; 2002.
10. Bezerra PRS, Schramm F, Schramm VB. A literature review on models for assessing corporate sustainability. *Int J Sustain Eng*. 2021;14(6):1306–18. <https://doi.org/10.1080/19397038.2021.1938291>.
11. Blom Invest Bank. *The Lebanese economy in 2018: time for drastic reforms*. Beirut: Blom Bank; 2018.
12. Boyatzis RE. *Transforming qualitative information: thematic analysis and code development*. Thousand Oaks, CA: Sage; 1998.
13. Braun V, Clarke V. (Mis) conceptualising themes, thematic analysis, and other problems with Fugard and potts' (2015) sample-size tool for thematic analysis. *Int J Soc Res Methodol*. 2016;19(6):739–43.
14. Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol*. 2006;3(2):77–101.
15. Budeanu A, Miller G, Moscardo G, Ooi CS. Sustainable tourism, progress, challenges and opportunities: an introduction. *J Clean Prod*. 2016;111:285–94. <https://doi.org/10.1016/j.jclepro.2015.10.027>.

16. Butler R. Measuring tourism success: alternative considerations. *Worldw Hosp Tour Themes*. 2022;14(1):11–9. <https://doi.org/10.1108/WHATT-10-2021-0133>.
17. Cardoso C. The contribution of tourism towards a more sustainable and inclusive society: key guiding principles in times of crisis. *Worldw Hosp Tour Themes*. 2020;12(6):679–89. <https://doi.org/10.1108/WHATT-07-2020-0065>.
18. Cresswell JW, Plano Clark VL. Designing and conducting mixed method research. 2nd ed. Thousand Oaks, CA: Sage; 2011.
19. Creswell JW, Poth CN. Qualitative inquiry and research design: choosing among five approaches. Thousand Oaks (CA): SAGE Publications; 2016. Available from: <https://doi.org/10.1177/1524839915580941>
20. Daou A, El Zeenni L, Hourani A, Talhouk SN. Ecotourism and grand challenges: a Gen-Z perspective in the context of Lebanon. In: *Tourist behaviour and the new Normal, volume II: implications for sustainable tourism development*. Cham: Springer Nature Switzerland; 2024. pp. 339–56.
21. Dias Á, Viana J, Pereira L. Barriers and policies affecting the implementation of sustainable tourism: the Portuguese experience. *J Policy Res Tour Leis Events*. 2024;1–19.
22. Dwyer L, Edwards D, Mistilis N, Roman C, Scott N. Destination and enterprise management for a tourism future. *Tour Manag*. 2009;30(1):63–74.
23. Dwyer L, Forsyth P, Dwyer W. Tourism economics and policy. 2nd ed. Channel View; 2020.
24. El Alam N, Stoffelen A, Bolderman L, Groote P. Tourism metagovernance and the role of the state: insights from post-blast Beirut. *Ann Tour Res*. 2024;109:103848. <https://doi.org/10.1016/j.jannals.2024.103848>.
25. El Zein A, Naimy V, Kordab M. The impact of tourism development on poverty alleviation: evidence from the Lebanese market. *Geoj Tour Geosites*. 2025;59(2):822–34.
26. Elgammal I, Ghanem M, Al-Modaf O. Sustainable purchasing behaviors in generation Z: the role of social identity and behavioral intentions in the Saudi context. *Sustainability*. 2024;16(11):4478.
27. Elkington J. Cannibals with forks: the triple bottom line of 21st century business. Stoney Creek, CT: New Society; 1997.
28. Farmaki A, Farmakis P. A stakeholder approach to CSR in hotels. *Ann Tour Res*. 2018;68:58–60.
29. Fernando S, Lawrence S. A theoretical framework for CSR practices: integrating legitimacy theory, stakeholder theory and institutional theory. *J Theor Acc Res*. 2014;10:149–78.
30. Freeman RE, Dmytryiev S. Corporate social responsibility and stakeholder theory: learning from each other. *Symphonya Emerg Issues Manag*. 2017;(1):7–15.
31. Freeman RE. Strategic management: a stakeholder approach. Boston (MA): Pitman Publishing; 1984.
32. Fugard AJ, Potts HW. Supporting thinking on sample sizes for thematic analyses: a quantitative tool. *Int J Soc Res Methodol*. 2015;18(6):669–84.
33. Giaccone SC, Bonacini E. New technologies in smart tourism development: the #iziTRAVELSicilia experience. *Tour Anal*. 2019;24(3):341–54.
34. Gilmore A, Carson D, Ascensão M. Sustainable tourism marketing at a world heritage site. *J Strateg Mark*. 2007;15(2–3):253–64.
35. Goel P. Triple bottom line reporting: an analytical approach for corporate sustainability. *J Finance Acc Manag*. 2010;1(1):27–42.
36. Gopalakrishnan K, Yusuf YY, Musa A, Abubakar T, Ambursa HM. Sustainable supply chain management: a case study of British aerospace systems. *Int J Prod Econ*. 2012;140(1):193–203.
37. Guest G, Bunce A, Johnson L. How many interviews are enough? An experiment with data saturation and variability. *Field Methods*. 2006;18(1):59–82.
38. Gutiérrez I, Alcaraz JM, Susaeta L, Suárez E, Pin JR. Managing sustainability for competitive advantage: evidence from the hospitality industry. *SSRN Electron J*. 2015. <https://doi.org/10.2139/ssrn.2598500>.
39. Hamati K. Lebanon. In: *Tourism governance: a critical discourse on a global industry*, vol 9. 2022. pp. 115.
40. Harmon W. The future of sustainability in the hospitality industry [Internet]. 2017 [cited 2025 May 13]. Available from: <https://www.bluecart.com/blog/the-future-of-sustainability-in-the-hospitality-industry>
41. Harrington RJ, Chathoth PK, Ottenbacher MC, Altinay L. Strategic management research in hospitality and tourism: past, present and future. *Int J Contemp Hosp Manag*. <https://doi.org/10.1108/ijchm-12-2013-0576>
42. Hassan V, Bellos G, Fawaz R. Transportation towards tourism sustainability: case study of Lebanon. *Athens J Tour*. 2021;8(3):177–92.
43. Hatipoglu B, Alvarez MD, Ertuna B. Barriers to stakeholder involvement in the planning of sustainable tourism: the case of the Thrace region in Turkey. *J Clean Prod*. 2016;111:306–17. <https://doi.org/10.1016/j.jclepro.2014.11.059>.
44. Hogrefe R, Bohnet-Joschko S. The social dimension of corporate sustainability: review of an evolving research field. *Sustainability*. 2022;15:3248.
45. Hull. Competitive sustainability: the intersection of sustainability and business success. *Sustainability*. 2022;14:16420.
46. International Tourism Partnership. ITP goals for 2030 unite the hotel industry for a sustainable future. 2018. <https://bit.ly/2RoeLQN>
47. Jones P, Comfort D. The COVID-19 crisis and sustainability in the hospitality industry. *Int J Contemp Hosp Manag*. 2020. <https://doi.org/10.1108/ijchm-04-2020-0357>.
48. Kabbara N, Ozgit H. Effectiveness of resource management of Lebanese ngos in response to COVID-19 and the Syrian crisis. *Humanit Soc Sci Commun*. 2023;10(1):1–11. <https://doi.org/10.1057/s41599-023-02261-1>.
49. Kaftan V, Kandalov W, Molodtsov I, Sherstobitova A, Strielkowski W. Socio-economic stability and sustainable development in the post-COVID era: lessons for the business and economic leaders. *Sustainability*. 2023;15:2876.
50. Kalargyrou V, Costen WM. Diversity management research in hospitality and tourism: past, present and future. *Int J Contemp Hosp Manag*. 2017. <https://doi.org/10.1108/IJCHM-09-2015-0470>.
51. Kasemsap K. Mastering employee turnover intention in the modern workforce. In: *Social issues in the workplace: breakthroughs in research and practice*. Hershey (PA): IGI Publishing/IGI Global; 2018. pp. 261–81.
52. Kataya A. The impact of COVID-19 pandemic on international and Lebanese tourism indicators. *Econ Appl Inf*. 2021;2:5–13.
53. Khan S, Akbar A, Nasim I, Hedvicakova M, Bashir F. Green finance development and environmental sustainability: a panel data analysis. *Front Environ Sci*. 2022;10:2134.
54. Khatter A, McGrath M, Pyke J, White L, Lockstone-Binney L. Analysis of hotels' environmentally sustainable policies and practices. *Int J Contemp Hosp Manag*. 2019;31:2394–410.

55. Khatter A, White L, Pyke J, McGrath M. Stakeholders' influence on environmental sustainability in the Australian hotel industry. *Sustainability*. 2021;13(3):1351.
56. Khizar HMU, Younas A, Kumar S, Akbar A, Poulouva P. The progression of sustainable development goals in tourism: a systematic literature review of past achievements and future promises. *J Innov Knowl*. 2023;8(4):100442. <https://doi.org/10.1016/j.jik.2023.100442>.
57. Kim YH, Barber N, Kim D-K. Sustainability research in the hotel industry: past, present, and future. *J Hosp Mark Manag*. 2019;28(5):576–620.
58. Kinyanjui DN. Adaptive responses to climate change: evaluating the implementation and effectiveness of water resource management strategies in selected hotels within the environs of lake Naivasha. *Int J Environ Clim Change*. 2019. <https://doi.org/10.9734/ijec/2019/v9i530115>.
59. Kronenberg K, Fuchs M. Aligning tourism's socio-economic impact with the united nations' sustainable development goals. *Tour Manag Perspect*. 2021;39:1–12.
60. Kularatne T, Wilson C, Månsson J, Hoang, Vincent, Lee B. Do environmentally sustainable practices make hotels more efficient? A study of major hotels in Sri Lanka. *Tour Manag*. 2019;71:213–25.
61. Ladki SM, Sadik MW. Factors affecting the advancement of the Lebanese tourism industry. *Islam and business*. Routledge; 2014. pp. 171–85.
62. Lockstone-Binney L, Ong F. The sustainable development goals: the contribution of tourism volunteering. In: *A sustainable tourism workforce*. London: Routledge; 2024. pp. 279–96.
63. López-Gamero MD, Pereira-Moliner J, Molina-Azorín JF, Tarí JJ, Pertusa-Ortega EM. Human resource management as an internal antecedent of environmental management: a joint analysis with competitive consequences in the hotel industry. *J Sustain Tour*. 2020;31(6):1293–314. <https://doi.org/10.1080/09669582.2020.1841216>.
64. Makdissi R, Attal L, Mekdessi S. Resilience and adaptation: navigating economic crisis in Lebanon – transformation of commercial company into industrial entity during uncertainty. *IOSR J Bus Manag*. 2023;25(8, Ser. 5):62–76.
65. Mammadov R. The importance of transportation in tourism sector. In: *Seventh Silk road international conference*. 2012; pp. 24–26.
66. Mekawy MA. A constructivist grounded theory investigation of businesses' concerns about public-private partnership responses toward COVID-19. *Tour Int Interdiscip J*. 2022;70(1):9–27.
67. Meneses D, Costa C, Ferreira FA, Eusébio C. A bibliometric overview of economic sustainability in tourism. 2024.
68. Miles M, Huberman AM. *Qualitative data analysis: an expanded sourcebook*. 2nd ed. Thousand Oaks (CA): Sage; 1994.
69. Miles MB, Huberman AM, Saldaña J. *Qualitative data analysis: a methods sourcebook*. 3rd ed. Thousand Oaks, CA: Sage; 2014.
70. Modica PD, Altinay L, Farmaki A, Gursoy D, Zenga M. Consumer perceptions towards sustainable supply chain practices in the hospitality industry. *Curr Issues Tourism*. 2020;23(3):358–75.
71. Mukhtorova N, Askarov Z, Battaglia A. Exploring smart tourism: lessons from abroad and opportunities for Uzbekistan. *YASHIL IQTISODIYOT VA TARAQQIYOT*. 2023;1(maxsus son).
72. Müller M, Vogel C. Eco-certification programs in Germany: an evaluation of their impact on the hotel industry. *J Environ Manag*. 2024;328:115019.
73. Nagendrakumar N, Alwis KNN, Eshani UAK, Kaushalya SBU. The impact of sustainability practices on the going concern of the travel and tourism industry: evidence from developed and developing countries. *Sustainability*. 2022;14:17046.
74. Namdar R, Karami E, Keshavarz M. Climate change and vulnerability: the case of MENA countries. *ISPRS Int J Geo Inf*. 2021;10(11):794. <https://doi.org/10.3390/ijgi10110794>.
75. Nıkadimovs O, Rodčenkova A. Hospitality industry during COVID-19 crisis: review of the current situation and sustainable post-crisis response. *Soc Integr Educ Proc Int Sci Conf*. 2021;6. <https://doi.org/10.17770/sie2021vol6.6432>.
76. OECD. *Rethinking tourism success for sustainable growth*. OECD tourism trends and policies. Paris: OECD Publishing; 2020.
77. Olya HS, Altinay L, Farmaki A, Kenebayeva A, Gursoy D. Hotels' sustainability practices and guests' familiarity, attitudes, and behaviours. *J Sustain Tour*. 2020.
78. Omair S, Khizar HMU, Majeed O, Iqbal MJ. Sustainability. Concept clarification and theory. *Corporate sustainability in Africa*. Cham: Springer Int Pub; 2023. pp. 375–404.
79. Onn CC, Mo KH, Radwan MK, Liew WH, Ng CG, Yusoff S. Strength, carbon footprint and cost considerations of mortar blends with high volume ground granulated blast furnace slag. *Sustainability*. 2019;11(24):7194. <https://doi.org/10.3390/su11247194>.
80. Ottaviani D, Demiröz M, Szemző H, De Luca C. Adapting methods and tools for participatory heritage-based tourism planning to embrace the four pillars of sustainability. *Sustainability*. 2023;15:4741.
81. Palguna IGR. Green human resource management practices in hospitality industry: a general review. *Int J Global Tour*. 2021. <https://doi.org/10.58982/injogt.v2i2.48>.
82. Parsakia K. Resource management strategies in the hospitality industry: balancing profit and sustainability. *J Resour Manag Decis Eng*. 2023;2(4):17–23.
83. Patton MQ. *Qualitative research and evaluation methods*. 3rd ed. Thousand Oaks, CA: Sage; 2002.
84. Peñaflores E. Triple bottom line practices on sustainability of selected hotels: basis for A proposed strategic plan. *Aloysian Interdiscip J Soc Sci Educ Allied Fields*. 2025;1(8):298–304.
85. Pereira V, Silva GM, Dias Á. Sustainability practices in hospitality: case study of a luxury hotel in Arrábida natural park. *Sustainability*. 2021;13(6):3164.
86. Petri L, Mikuli D. Protected areas and tourism development on Croatian islands: Coexistence or divergence? *WIT Trans Ecol Environ*. 2012;166:29–40.
87. Qiu H, Fan DXF, Lyu J, Lin PMC, Jenkins CL. Analyzing the economic sustainability of tourism development: evidence from Hong Kong. *J Hosp Tour Res*. 2019;43(2):226–48. <https://doi.org/10.1177/1096348018777046>.
88. Raad NG. A strategic approach to tourism development barriers in Iran. *J Tour Hosp*. 2019;8:410. <https://doi.org/10.4172/2167-0269.1000410>.
89. Rasoolimanesh SM, Ramakrishna S, Hall CM, Esfandiar K, Seyfi S. A systematic scoping review of sustainable tourism indicators in relation to the sustainable development goals. *J Sustain Tour*. 2023;31(7):1497–517.
90. Riensche M, Carillo A, Garcia-Frapoli E, Monteno-Casasola P, Tello-Díaz C. Private over public interests in regional tourism governance: a case study in Costalegre. *Mexico Sustain*. 2019;11:1760.

91. Rogers K, Hudson B. The triple bottom line: the synergies of transformative perceptions and practices of sustainability. *OD Pract.* 2011;43(4):3–9.
92. Runtuk JK, Ng PK, Ooi SY, Purwanto R, Nur Chairat AS, Ng YJ. Sustainable growth for small and medium-sized enterprises: interpretive structural modeling approach. *Sustainability.* 2023;15:4555.
93. Santhanam N, Kamalanabhan TJ, Dyaram L, Ziegler H. Impact of human resource management practices on employee turnover intentions. *J Indian Bus Res.* 2017. <https://doi.org/10.1108/jibr-10-2016-0116>.
94. Scheyvens R. Linking tourism to the sustainable development goals: a geographical perspective. *Tour Geogr.* 2018;20(2):341–2.
95. Schwartz K, Tapper R, Font X. A sustainable supply chain management framework for tour operators. *J Sustain Tour.* 2008;16(3):298–314.
96. Scott D, Gössling S. A review of research into tourism and climate change. *Ann Tour Res.* 2022;95:103409. <https://doi.org/10.1016/j.annals.2022.103409>.
97. Siddique MNEA, Nor SM, Senik ZC, Omar NA. Corporate social responsibility as the pathway to sustainable banking: a systematic literature review. *Sustainability.* 2022;15(3):1807.
98. Sigala M, Tourism. COVID-19: impacts and implications for advancing and resetting industry and research. *J Bus Res.* 2020;117:1–22. <https://doi.org/10.1016/j.jbusres.2020.06.015>.
99. Simpson MC, Wallis A. Alternative interpretations: exceptional circumstances and the Bluestone development in Pembrokeshire Coast National park. *Tour Hosp Plan Dev.* 2008;5(3):203–14.
100. Simpson MC. Green tourism planning: Triple bottom line sustainability—rhetoric or reality? A case study of Bluestone development. *Eur J Tour Res.* 2008;1(2):147–152.2021;27(2):5–13. <https://doi.org/10.35219/eai15840409185>
101. Simpson MC. Using discourse analysis to explore the achievability of triple bottom line sustainability: the case of the Bluestone holiday Village. In: academic renewal: innovation in leisure and tourism theories and methods. *LSA Publ.* 2007;97:1–14.
102. Sobaih AEE, Hasanein A, Elshaer I. Influences of green human resources management on environmental performance in small lodging enterprises: the role of green innovation. *Sustainability.* 2020;12(24):10371. <https://doi.org/10.3390/su122410371>.
103. Sodikov M. Technological leadership and structural adaptation of industries of the Russian federation in the context of tourism. *Econ Dev Anal.* 2025;3(1):162–8.
104. Spangenberg JH. Economic sustainability of the economy: constructs and indicators. *Int J Sustain Dev.* 2005;8(1/2):47–64. <https://doi.org/10.1504/IJSD.2005.007374>.
105. Spradley JP. The ethnographic interview. New York: Holt, Rinehart & Winston; 1979.
106. Susanto EH, Rofaída R, Senen SH. Green human resources management in hospitality industries: a systematic literature review. *Eur J Hum Resour Manag Stud.* 2022;5(4). <https://doi.org/10.46827/ejhrms.v5i4.1212>.
107. Torres-Delgado A, Palomeque FL. The growth and spread of the concept of sustainable tourism: the contribution of institutional initiatives to tourism policy. *Tourism Manag Perspect.* 2012;4:1–10.
108. Trupp A, Dolezal C. Tourism and the sustainable development goals in Southeast Asia. *ASEAS.* 2020;13(1):1–16.
109. United Nations Global Compact. Global goals for people and planet [Internet]. 2018 [cited 2018 Dec 3]. Available from: <https://bit.ly/2vzbzNN>
110. United Nations World Tourism. Organization (UNWTO). Annual report. 2017.
111. United Nations. About the sustainable development goals [Internet]. 2018 [cited 2025 May 11]. Available from: <https://bit.ly/2HjQmD>
112. Vachon S. Green supply chain practices and the selection of environmental technologies. *Int J Prod Res.* 2007;45(18–19):4357–79.
113. Verma VK, Chandra B. Sustainability and customers' hotel choice behaviour: a choice-based conjoint analysis approach. *Environ Dev Sustain.* 2018;20(3):1347–63.
114. Vieira I, Rodrigues A, Alén E, Sousa N. Challenges of sustainable tourism management in a small historical town from the perspective of accommodation managers. *RPER.* 2024;(69):57–78.
115. Waligo VM, Clarke J, Hawkins R. Implementing sustainable tourism: a multi-stakeholder involvement management framework. *Tour Manag.* 2013;36:342–53.
116. Wang ZH, Zhao X, Yang J, Song J. Cooling and energy saving potentials of shade trees and urban lawns in a desert City. *Appl Energy.* 2016;161:437–44.
117. Wijesundara N. Corporate sustainability practices of hotel industry: a systematic literature review. *Lond J Res Manage Bus.* 2023;23(5):43–53.
118. World Bank. Lebanon economic monitor. Spring 2023: the normalization of crisis is no road for stabilization. 2023. <https://www.worldbank.org/en/country/lebanon/publication/lebanon-economic-monitor-spring-2023-the-normalization-of-crisis-is-no-road-for-stabilization>
119. World Tourism Organisation, United Nations Global Compact Network Spain. The tourism sector and the sustainable development goals: responsible tourism, a global commitment [Internet]. 2016 [cited 2018 Dec 2]. Available from: <https://bit.ly/2RRytng>
120. World Tourism Organization, United Nations development programme. Tourism and the sustainable development goals – Journey to 2030: highlights [Internet]. 2017 [cited 2025 May 23]. Available from: <https://www.e-unwto.org/doi/book/https://doi.org/10.18111/9789284419401>
121. Wu L, Jin S. Corporate social responsibility and sustainability: from a corporate governance perspective. *Sustainability.* 2022;14:15457.
122. Yacoub L. Toward sustainable hospitality: an examination of sustainable hotel practices in Lebanon. *Navigating the intersection of Business, sustainability and technology.* Singapore: Springer Nature; 2024. pp. 1–23.
123. Yin RK. Case study research: design and methods. 4th ed. Thousand Oaks (CA): SAGE; 2009.
124. Žak A. Triple bottom line concept in theory and practice. *Soc Responsib Organ Dir Chang.* 2015;387(1):251–64.

125. Zhang J. Analysis of strategic human resource management practices. *Front Bus Econ Manag*. 2023. <https://doi.org/10.54097/fbem.v8i3.7872>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.